

Keywords: scaling laws • inference-time compute • chain-of-thought • Bayesian analysis

Scaling of Capability and Efficiency

Joint Quantitative Laws for Solve Rate and Token Cost in Large Reasoning Models

By (authors not specified in pre-print metadata)

arXiv:2609.27166 • September 2026

Using hierarchical Bayesian models applied to the DeepSeek-R1-Distill family, this paper shows that solve probability decays exponentially with problem hardness while the characteristic scale grows as a power law with model parameters—and that among solved instances, output length grows as a power law with instance size, providing the first joint quantitative account of capability and efficiency scaling at inference time.

AT A GLANCE

Problem: Capability and efficiency in chain-of-thought reasoning both depend on problem hardness and model size, but their joint scaling behaviour has not been quantitatively characterized.

Why it matters: Without joint scaling laws, predicting compute requirements for a given problem difficulty and model size is guesswork.

Method: Hierarchical Bayesian models fit to the DeepSeek-R1-Distill family across four algorithmic problem classes with controlled hardness.

Result: Exponential capability decay with power-law characteristic scale; power-law token cost growth among correctly solved instances.

Evidence: 1.5B–70B parameter DeepSeek-R1-Distill models on arithmetic, multiplication, sorting, and path-finding.

The Big Picture

Large reasoning models (LRMs) achieve state-of-the-art results by generating extended chain-of-thought traces before producing final answers. The amount of computation consumed at inference time—measured in output tokens—grows substantially with problem difficulty, and harder problems are solved by larger models. Both capability (solve rate) and efficiency (tokens per correct solution) thus depend on at least two variables: problem hardness and model size.

Despite the obvious importance of understanding this dependency, it has not been characterized in the form of

quantitative scaling laws. Practitioners must therefore guess how many tokens a given model will require to solve problems of a given difficulty, and how much larger a model must be to solve problems that the current model cannot.

This paper provides the first joint quantitative account of capability and efficiency scaling at inference time, using controlled algorithmic problem classes where hardness is operationalized as instance size and correctness is verified exactly.

Why Existing Approaches Fall Short

Average accuracy evaluations aggregate performance across problem difficulties, masking the exponential decay with hardness and conflating easy-instance dominance with true capability on hard instances.

Single-model comparisons at fixed compute budgets cannot reveal how model scale shifts the capability-efficiency frontier.

Simple regression on token counts ignores the joint structure of success and token cost, treating failed attempts identically to efficient successes.

Prior scaling laws (Chinchilla, etc.) address training compute and not inference-time token generation conditioned on problem difficulty, leaving the test-time compute dimension uncharacterized.

Core Mechanism

Hierarchical Bayesian Model. The paper fits a generative model to solve indicators and token counts across models and problem instances. The model has three layers:

1. **Instance level:** binary solve indicator and token count per instance per model.
2. **Class level:** partial pooling of scaling exponents across four problem classes.
3. **Model level:** power-law relationship between model size and scaling parameters.

Problem Classes. Four algorithmic tasks with exact verifiers: arithmetic computation, digit multiplication, list sorting, and path-finding. Hardness is controlled by instance size n (digits, list length, or graph nodes).

Model Family. DeepSeek-R1-Distill at 1.5B, 7B, 14B, 32B, and 70B parameters—a controlled family where differences arise from model scale alone.

Technical Formulation

Solve probability at instance size n and model parameter count m :

$$p(n, m) = p_0(m) \cdot \exp\left(-\frac{n}{\lambda(m)}\right)$$

Characteristic scale as a power law in model parameters:

$$\lambda(m) = \alpha \cdot m^\beta$$

Conditional output length (correctly solved instances):

$$\mathbb{E}[L \mid \text{solved}, n, m] = \gamma(m) \cdot n^\delta$$

Hierarchical priors (partial pooling across problem classes $k = 1, \dots, K$):

$$\beta_k \sim \mathcal{N}(\mu_\beta, \sigma_\beta^2), \quad \delta_k \sim \mathcal{N}(\mu_\delta, \sigma_\delta^2)$$

All parameters are jointly estimated by MCMC, yielding posterior distributions over scaling exponents β and δ per class.

Learning or Inference Procedure

1. For each (model, problem class, instance size): generate up to R attempts; record solve indicator and output token count.
2. Fit the hierarchical Bayesian model to the full dataset using MCMC (Stan or equivalent).
3. Report posterior means and 90% credible intervals for $\alpha, \beta, \gamma, \delta$ per class.
4. Use fitted model to predict solve probability and expected token cost for new (model size, instance size) pairs.

What the Guarantee Says

The hierarchical model provides posterior uncertainty over scaling exponents. Within the linear-arithmetic and combinatorial-algorithmic problem classes studied, exponential solve-rate decay with power-law characteristic scale is a consistent structural finding, not an artifact of a single model or problem class. Cross-class partial pooling provides regularization that stabilizes estimates at large difficulty where data is sparse.

Experimental Findings

- **5 models:** 1.5B to 70B DeepSeek-R1-Distill
- **4 problem classes:** arithmetic, digit multiplication, list sorting, path-finding
- Solve probability fits exponential decay with hardness ($R^2 > 0.92$) across all classes

- Characteristic scale exponent $\beta \approx 0.3$ – 0.5 by class; path-finding has the steepest decay
- DeepSeek-R1-Distill-70B achieves approximately $10\times$ larger characteristic scale than the 1.5B model on arithmetic
- Output length exponent $\delta \approx 1.0$ – 1.5 : near-linear to super-linear growth with instance size
- Efficiency degrades faster than capability with hardness: diminishing returns to scale on the hardest instances

Ablations and Interpretation

Problem class differences: Path-finding shows the steepest capability decay, suggesting it requires qualitatively more extensive search. Arithmetic shows the shallowest, consistent with near-algorithmic solutions.

Pooling vs. no pooling: Hierarchical pooling across classes stabilizes estimates at large difficulty; without pooling, estimates at the hardest difficulty levels have large variance.

Efficiency vs. capability trade-off: The power-law exponent $\delta > 1$ for output length implies that even when a larger model can solve harder instances, it pays a super-linear token cost. Compute-optimal scaling must balance capability and efficiency gains.

Implications: Given model size m and a target solve probability p^* on instances of size n^* , the minimum required model size is: $m^* = (\lambda^{-1}(-n^*/\ln p^*)/\alpha)^{1/\beta}$ —a prediction the framework makes interpretable and calibrated.

Reference

Scaling of Capability and Efficiency at Inference Time in Large Reasoning Models. arXiv:2609.27166, September 2026.

<https://arxiv.org/abs/2609.27166>